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SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

Code of Conduct:

I SORNAVARSHINI (1965260025) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Sornavarshini M.

Criteria	Weightage (%)	Q1 10marks	Q2 10marks	Q3 10marks
Concept Identification	20%	2	2	2
Concept Mapping Structure	25%	2	2	2
Linking & Relationships	20%	2	2	2
Application & Analysis	20%	2	2	2
Explanation & Clarity	15%	1	1	1

Total Marks 27 /30

Signature of the Faculty

CONCEPT MAPPING QUESTIONS

1. A partially completed concept map is given below:

Application → Presentation → _____ → Transport → Network → _____ → Physical.

Analyze the missing layers and explain how their functions are essential for reliable communication.

Analysis		concepts to explain processes (e.g., encapsulation, error control) with strong reasoning	correct but lacks depth in analysis	application with weak explanation	apply concepts correctly
Explanation & Clarity	15%	Provides a clear, concise, and well-structured explanation supporting the concept map	Explanation is understandable with minor gaps	Explanation is unclear or incomplete	No clear or meaningful explanation

ERROR CONTROL

Data link Layer

Error Detection

- Uses techniques like CRC (cycle Redundancy check) Parity bits, checksums, etc.
- Detects errors in the frame during transmission.
- If an error is found, the frame is discarded or marked for retransmission.

Transport Layer

Error Correction

- Uses techniques like ARQ (Automatic Repeat request) sequence numbers and error-correcting codes.
- Corrects errors that are detected (or requests retransmission if needed)
- Ensures end-to-end reliable delivery of data.

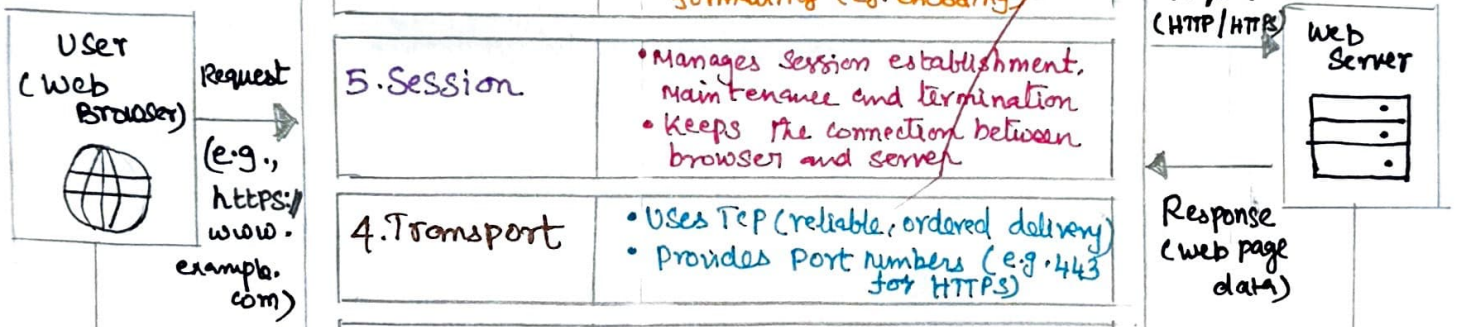
RESULT:

The Data Link Layer detects errors early at the frame level, and the Transport Layer corrects them (or ensures retransmission), working together to provide reliable and accurate data transmission from source to destination.

Web Browsing Using OSI Layers

OSI Layers and Their Role in Web Browsing

7. Application	• Provides web services (HTTP/HTTPS) • Handle data format (HTML, JSON etc.,)
6. Presentation	• Encrypts data (SSL/TLS for HTTPS) • Handles data compression and formatting (eg. encoding)
5. Session	• Manages session establishment, maintenance and termination • Keeps the connection between browser and server
4. Transport	• Uses TCP (reliable, ordered delivery) • Provides port numbers (e.g. 443 for HTTPS)
3. Network	• Handle logical addressing (IP address) • Route data across network (routing)
2. Data Link	• Frames data and user MAC addresses. • Detects/corrects errors in the local network (eg Ethernet/Wi-Fi)
1. Pysical	• Transmits raw bits over the medium (cables/wi-fi) • Handles signals, voltages and hardware.



Result

The browser receives the web page, which is rendered and displayed to the user.

How Failure in one Layer Affects the Entire Process

7. Application Layer Failure
- website not found / cannot load.
- No web page is displayed.

6. Presentation Layer Failure
- SSL/TLS failure → Security error.
- Garbled or unreadable data.

5. Session Layer Failure
- Session cannot be established
- Frequent logouts/connection drops.

4. Transport Layer Failure
- No reliable data delivery (Packet Loss).
- Page loads incompletely or times out.

3. Network Layer Failure
- Incorrect routing/unreachable Server.
- "Site can't be reached" error.

2. Data Link Layer Failure
- Frame errors/corrupted data.
- Intermittent disconnections.

1. Physical Layer Failure
- No Signal/broken cable/wifi issue
- Entire connection Lost.

Overall Impact

Since each Layer depends on the Previous one, a failure in any Single Layer breaks the chain, causing the web page to fail to load or display correctly.

Network Issue – Concept Map

Physical Layer (✓ Working)

- Cable, Switch and Physical connection are fine.
- Link is up.

Data Link Layer (✓ Working)

- Frames are sent and received correctly.
- MAC addressing and error detection work.
- Local network (LAN) is fine.

Network Layer (✗ Failing)

- Incorrect IP addressing, routing problems or no route to destination.
- Packets cannot move beyond and local network.

Transport Layer (✗ Failing)

- Web services (HTTP/HTTPS) cannot be reached.
- Browser shows "site can't be reached" or times out.

Application Layer (✗ Failing)

- Cannot establish connection (eg TCP/UDP)
- No end-to-end delivery
- Timeout / connection errors may occur.

Analysis of the Problem

1. Physical Layer (✓) : No issue. Cable, Switch and link are working.
2. Data Link Layer (✓) : Frames are sent/received correctly. MAC addressing and error detection work.
3. Network Layer (✗) : Fails due to incorrect IP addresing routing problems or no route to designation. Packets cannot move beyond local Network.
4. Transport Layer (✗) : Cannot establish connection (TCP/UDP). No end-to-end delivery.
5. Application Layer (✗) : Web Series (HTTP/HTTPS) cannot be reached. Browser show "Site can't be reached" or time out

How the Concept Map Helps in Troubleshooting

Locate the Fault.

↓

Check Dependent Layers.

↓

Focus on Relevant Tests.

↓

Avoid Unnecessary checks

↓

Resolve and verify.

Conclusion.

The concept map helps in quickly identifying the faulty layer (Network Layer) and show how the failure effect the upper layer guiding the troubleshooting process.